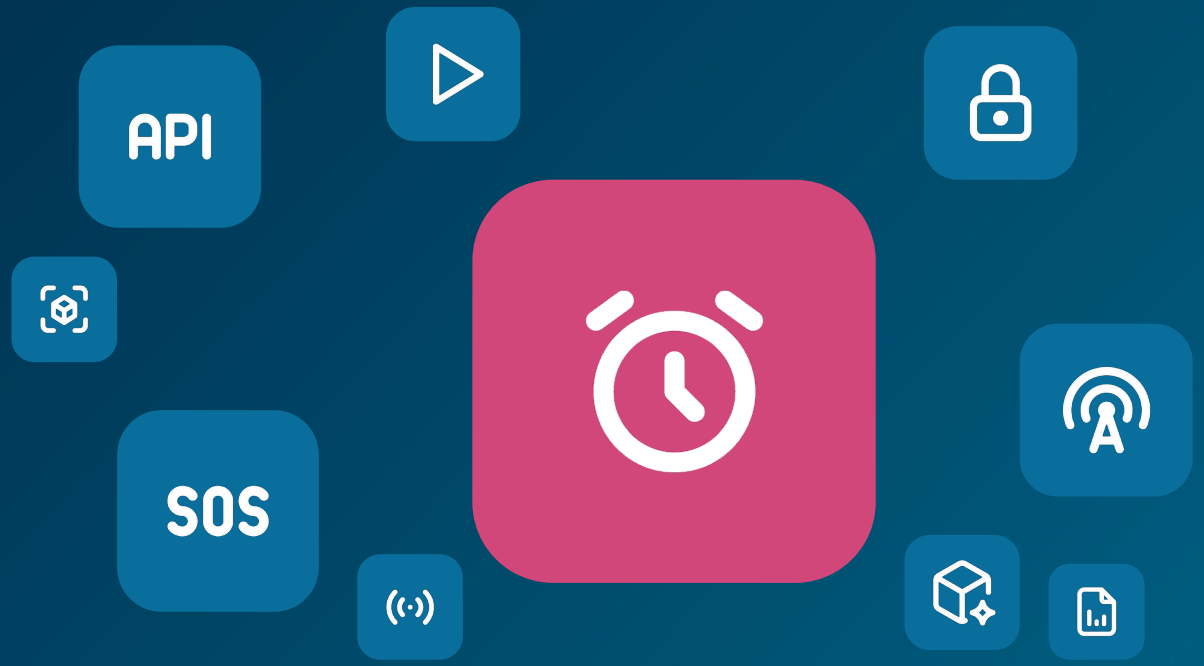
5G MAG The Media Connectivity Association



OVERVIEW AND UPDATES · CONNECTED MEDIA PRODUCTION

Time-Sensitive Communications (TSC)

Deterministic, low-jitter transport for tightly synchronised professional media

3GPP

OCTOBER 2026 – www.5g-mag.com

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About the project

The 5G system as a bridge between wired Time-Sensitive Networking segments

TS 23.501

3GPP system architecture

The 5G system modelled as a TSN bridge, NW-TT and DS-TT, and time synchronisation.

TS 23.502

3GPP procedures

The procedures of Time-Sensitive Communications.

TS 29.565

3GPP stage 3

The services of the Time Sensitive Communication and Time Synchronization Function (TSCTSF). IEEE 802.1 TSN standards are referred to throughout.

FIND OUT MORE **Technical analysis** **Standards**

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Control plane

How 5G carries deterministic traffic and shared timing between wired TSN segments

AT THE UPF

NW-TT

The Network-side TSN Translator at the UPF. It terminates the wired TSN network and holds the bridge management information.

IN THE 5G CORE

TSCTSF

The network function through which deterministic QoS and time synchronisation are requested. An AF outside the trust domain reaches it through the NEF.

IN THE TSN NETWORK

TSN CNC

Computes schedules across the 5G bridge with ordinary TSN tooling; the 5G system reports its bridge capabilities to it.

The 5G system is modelled as one or more virtual TSN bridges, so wired TSN tooling can schedule across the wireless hop.

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Specifications in scope

The standards that 5G TSC interworks with and the media workflow it carries

3GPP service requirements and system architecture

- TS 22.104
- TR 22.804
- TS 23.501
- TS 23.502
- TR 23.700-25
- TS 29.565
- TS 29.522

IEEE TSN Time-Sensitive Networking

- 802.1AS timing and synchronisation
- 802.1Q bridges
- 802.1BA Audio Video Bridging
- 802.1CB frame replication

SMPTE IP media production

- ST 2110-10 timing
- ST 2110-20 video
- ST 2110-21 traffic shaping
- ST 2110-30 audio
- ST 2110-40 ancillary data

TSC in a media context usually runs over a private 5G deployment, so it builds on Non-Public Networks.

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How it grew

What each 3GPP release added to Time-Sensitive Communications

Release 16

The 5G system as a TSN bridge, with DS-TT and NW-TT. gPTP is forwarded transparently, with residence-time correction, and TSN QoS is mapped to 5G QoS.

Release 17

The TSCTSF, and time synchronisation generalised to several IEEE 802.1AS and IEEE 1588 clock roles. Deterministic QoS and time-sync control are exposed to AFs through the NEF, without a full TSN bridge.

Release 18 and later

Further deterministic networking, survivability and time synchronisation accuracy enhancements. The scope is to be confirmed against the 3GPP work plan.

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